

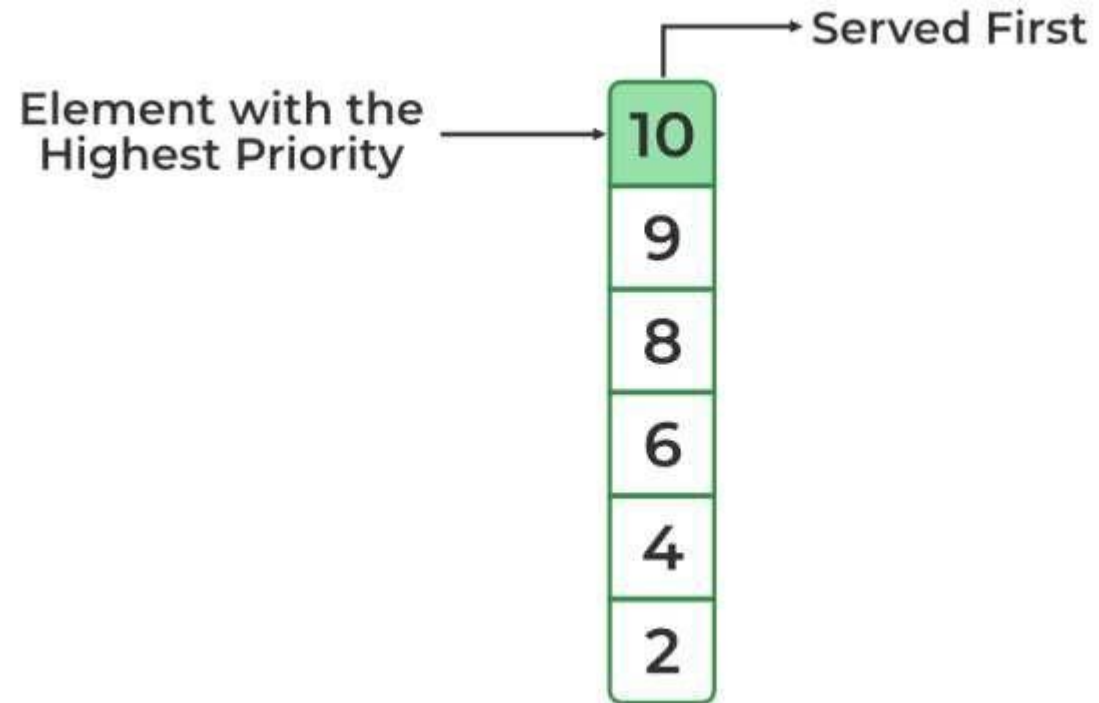
23CS201 - Data Structures and Algorithms using C++

Study Materials

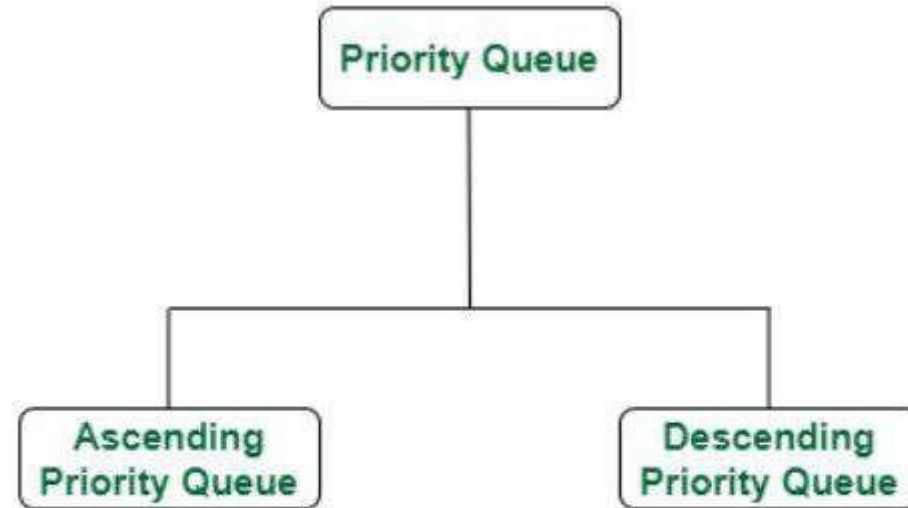
Priority Queue:

- A Priority Queue is a special type of queue where each element is assigned a priority, and the element with the highest priority is dequeued first.
- **Types of Priority Queue:**
- Min Priority Queue: The element with the lowest value is given the highest priority.
- Max Priority Queue: The element with the highest value is given the highest priority.
- Implementation:
- Priority queues can be implemented using arrays, linked lists, or heaps (binary heaps are common for efficient operations).

Representation of Priority Queue:



Types of Priority Queue:



Types of Priority Queue

Queue Methods:

- In C++, the priority queue class provides various methods to perform different operations on a queue.

Methods	Description
<code>push()</code>	inserts the element into the priority queue
<code>pop()</code>	removes the element with the highest priority
<code>top()</code>	returns the element with the highest priority
<code>size()</code>	returns the number of elements
<code>empty()</code>	returns <code>true</code> if the priority_queue is empty

Difference between Queue and Priority Queue:

Priority Queue	Queue
Process the elements with the highest priority	No priority Exists
No principle applied	Follows FIFO principle
If more than one element with the same priority exists, the order of queue is taken	No such process undergoes

Fundamental Operations:

- The priority queue supports certain basic activities.
- Insert(item, priority): Inserts an item with the specified priority into the priority queue.
- Get Highest Priority(): Returns the highest priority item.
- Delete Highest Priority(): Removes the highest priority item.
- In addition to the aforementioned actions, the standard queue operations such as isEmpty (), isFull (), and peek ().

Properties of Priority Queue:

- So, a priority Queue is an extension of the queue with the following properties.
- Every item has a priority associated with it.
- An element with high priority is dequeued before an element with low priority.
- If two elements have the same priority, they are served according to their order in the queue.

Enqueue & Dequeue:

Priority Queue

Insertion

Enqueue (1,50)	1			Queue
	50			Priority
Enqueue (2,40)	1	2		Queue
	50	40		Priority
Enqueue (3,30)	1	2	3	Queue
	50	40	30	Priority

Deletion

Dequeue		2	3	Queue
		40	30	Priority
Dequeue			3	Queue
			30	Priority

Applications:

1. Task Scheduling:

- Operating systems use queues to manage tasks in multi-tasking environments.

2. Data Buffering:

- Queues are used in buffers for IO devices (e.g., keyboards, printers).

3. Breadth-First Search (BFS):

- Used in graph traversal algorithms like BFS.

4. Simulation Systems:

- Queues are used in simulation models to handle customer service systems, traffic management, etc.

Algorithm :

- POP(HEAD):
 - Step 1: Set the head of the list to the next node in the list. $HEAD = HEAD \rightarrow NEXT$.
 - Step 2: Free the node at the head of the list
 - Step 3: End
- PEEK(HEAD):
 - Step 1: Return $HEAD \rightarrow DATA$
 - Step 2: End